QMSS G5063 Final Project Proposal

Title: Mapping Restaurant Quality in New York City: Inspections, Neighborhood Patterns, and Review Language

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Abstract

We propose to build an interactive data visualization project that examines restaurant quality and dining experiences across New York City. Our project will combine official NYC restaurant inspection records, geographic boundary data, and public restaurant review data to explore how restaurant inspection outcomes vary across boroughs, neighborhoods, and cuisine types, and whether public review language aligns with official health inspection results.

The project will focus on three related questions. First, how do restaurant grades, inspection scores, and health code violations vary across different parts of New York City? Second, are some cuisine categories or neighborhoods associated with systematically better or worse inspection outcomes? Third, does the language used in public reviews reflect patterns found in official inspection data, such as lower grades, frequent critical violations, or higher quality ratings?

Our final product will be an interactive Streamlit website that combines maps, charts, and simple text analysis. This project is designed to include interactive visualization and at least two specialized visualization types. In our case, these will be **geospatial visualization** and **text analysis**. The course guidelines explicitly encourage projects with geographic identifiers and text, and recommend building the final output as a publicly accessible interactive website.

Data Sources

We plan to combine the following datasets:

1. Official NYC Restaurant Inspection Data

Our primary dataset will be the **DOHMH New York City Restaurant Inspection Results** from NYC Open Data. This dataset includes restaurant identifiers, names, boroughs, addresses, cuisine descriptions, inspection dates, scores, grades, and violation information. This will serve as the core analytical table for the project.

Key fields we expect to use:

- restaurant ID
- restaurant name
- borough
- building, street, ZIP code
- cuisine description
- inspection date
- score
- grade
- violation code and description

- critical flag
- latitude and longitude, if available

Source:
https://data.cityofnewyork.us/Health/DOHMH-New-York-City-Restaurant-Inspection-Results/43nn-pn8j/about_data

2. Geographic Boundary Data
To support mapping, we will use NYC geographic boundary files from NYC Open Data. We are currently considering:

- **Borough Boundaries** for a simpler and more stable map layer
- **Neighborhood Tabulation Areas (NTAs)** for a more detailed neighborhood-level analysis

If time is limited, we will prioritize borough boundaries first and add NTAs only if the merge and mapping process remains manageable.
Source:
https://data.cityofnewyork.us/City-Government/Borough-Boundaries/gthc-hcne/about_data
(The boundaries of the five major regions are most suitable for making a simple and stable choropleth map.)
https://data.cityofnewyork.us/City-Government/2020-Neighborhood-Tabulation-Areas-NTAs-/9nt8-h7nd/about_data
(The neighborhood level boundary is suitable for more detailed spatial analysis.)

3. Public Restaurant Review Data
To incorporate text analysis, we plan to use one public restaurant review dataset, likely a Kaggle dataset focused on NYC restaurants. We want to keep this part simple and manageable, so we will use only one review source rather than merging several platforms at once.
Fields we hope to use:

- restaurant name
- rating
- price level
- cuisine or category
- review text
- location information

This data structure fits the course goals well because the project combines a large dataset, geographic information for mapping, and text for text analysis.
Source:
https://www.kaggle.com/datasets/rayhan32/trip-advisor-newyork-city-restaurants-dataset-10k
https://www.kaggle.com/datasets/beridzeg45/nyc-restaurants

Research Questions
Our project will address the following questions:

1. **How do restaurant inspection outcomes vary across New York City?**
   We will compare inspection grades and scores across boroughs and, if feasible, neighborhoods.
2. **Are some cuisine categories or neighborhoods associated with systematically better or worse inspection outcomes?**
   We will examine whether certain cuisine types or areas are consistently associated with higher inspection scores, lower grades, or more frequent critical violations.
3. **Do public reviews align with official inspection results?**
   We will explore whether restaurants with worse inspection outcomes also receive more negative review language or lower ratings.

Planned Visualizations and Analyses
Following the project expectations, we plan to include both static and interactive visualizations in a Streamlit app.
Geospatial Visualizations
- Interactive borough map showing average inspection scores or grade distributions
- Neighborhood choropleth map if NTA-level analysis is feasible
- Restaurant point map showing restaurant locations, colored by grade or score

Standard Visualizations
- Bar charts comparing average inspection scores across boroughs and cuisines
- Stacked bar charts showing grade distributions by borough or cuisine
- Time trend line charts showing how inspections or average scores change over time
- Top violation charts displaying the most common violation types

Text Analysis Visualizations
- Word frequency charts for positive versus negative review groups
- Keyword comparison plots for restaurants with better and worse inspection outcomes
- Possibly a simple sentiment comparison across boroughs or cuisines if the text data quality is good enough

Tools and Libraries
- **Pandas**
- **Plotly** or **Altair**
- **GeoPandas** and **Folium** or **PyDeck** (maps)
- **Streamlit** (website)
- **spaCy** or basic NLP tools (simple text analysis)

Why This Version Is Feasible
First, the official NYC inspection data already contains many of the variables needed for strong descriptive analysis, so the project does not depend entirely on difficult web scraping or API collection. Second, borough-level mapping can produce meaningful results even if more detailed

neighborhood-level analysis becomes too complicated. Third, text analysis can remain simple by using one review dataset and basic word-frequency or sentiment summaries rather than advanced modeling.

Additional Datasets We May Need

Besides the three main data categories, there are a few useful supporting datasets or fields that may be needed:

- **ZIP code to borough/neighborhood crosswalk**, in case geographic joins are messy
- **Restaurant geolocation fields** or a geocoding-ready address field
- **Date standardization fields**, because inspection data may contain multiple records per restaurant over time
- **Cuisine category cleaning table**, since cuisine labels can be inconsistent

Potential Sources:

1. Zip code:

https://data.cityofnewyork.us/Health/Modified-Zip-Code-Tabulation-Areas-MODZCTA-/pri4-ifjk/about_data

2. Demographics:

https://data.cityofnewyork.us/City-Government/Census-Demographics-at-the-Neighborhood-Tabulation/rnsn-acs2/about_data

Potential Problems With the Current Idea

1. Restaurant matching across datasets may fail

The biggest problem for our project is that the official inspection dataset and the review dataset may spell names differently, use different addresses, include chain locations differently, or refer to restaurants that have closed or changed names. If matching fails, the text-analysis part could become much weaker than expected. To address the problem, we would keep one review source only, and be ready to do a limited merge using restaurant name + borough + partial address rather than trying to match everything.

2. The inspection data is not one row per restaurant

Inspection datasets often contain repeated inspections and repeated violation rows. This means we will need to decide whether our main unit is:

- each inspection record,
- each restaurant's most recent inspection,
- or restaurant-level averages over time.

For the inspection-based analyses, we will define the unit of analysis as the restaurant location, using each restaurant's most recent inspection to avoid overcounting repeated inspections. For the public review map, we will aggregate review measures to the borough or neighborhood level.

3. Neighborhood-level mapping may be harder than borough-level mapping

Borough-level analysis is easy and clean. Neighborhood-level analysis is more interesting, but it requires reliable geographic joins and possibly coordinate cleaning. That may become time-consuming. To address this problem, we will make borough-level mapping as our guaranteed version, and treat NTA mapping as an optional upgrade.